

Visualizing the regional patterns of two crises: The COVID-19 outbreak and decreasing MSME sales during three different phases of 2020 in Korea

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journals.sagepub.com/home/epn**Donghyun Kim** 

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Abstract

The coronavirus disease 2019 (COVID-19) pandemic has caused various socioeconomic problems as well as deaths. The featured graphic depicts the COVID-19 outbreak and the accompanying decrease in sales of micro, small, and medium enterprises (MSMEs) from week 2 to week 53 in 2020 as well as its regional patterns during three different phases in Korea. The geovisualization shows that the MSMEs' economic crisis occurred during the COVID-19 outbreak period. The COVID-19 crisis witnessed the emergence of pandemic management strategies, including mitigation and reinforcement of lockdowns repeatedly, which weakened the effectiveness of policies for controlling the spread of infectious diseases. This research sheds further light on the prioritization of policy-related decision making for controlling infectious diseases by utilizing economic shutdowns.

Keywords

COVID-19, economic crisis, lockdown

In Korea, the confirmation of the first case of the coronavirus disease 2019 (COVID-19) on January 20, 2020, was followed by 60,740 cumulative confirmed cases and 900 deaths in 1 year (Korea Disease Control and Prevention Agency, 2020). During the current COVID-19 pandemic, many national governments have used lockdowns to control the spread of the disease and to authorize relief packages in response to emergency economic shutdowns. Although the COVID-19 disease is a global pandemic, there have been several regional disparities (Chen et al., 2021; Zhai et al., 2021); thus, any analysis of its impact on micro, small, and medium enterprise (MSME) sales will also have to consider the influence of regional differences. Therefore, this study compares regional

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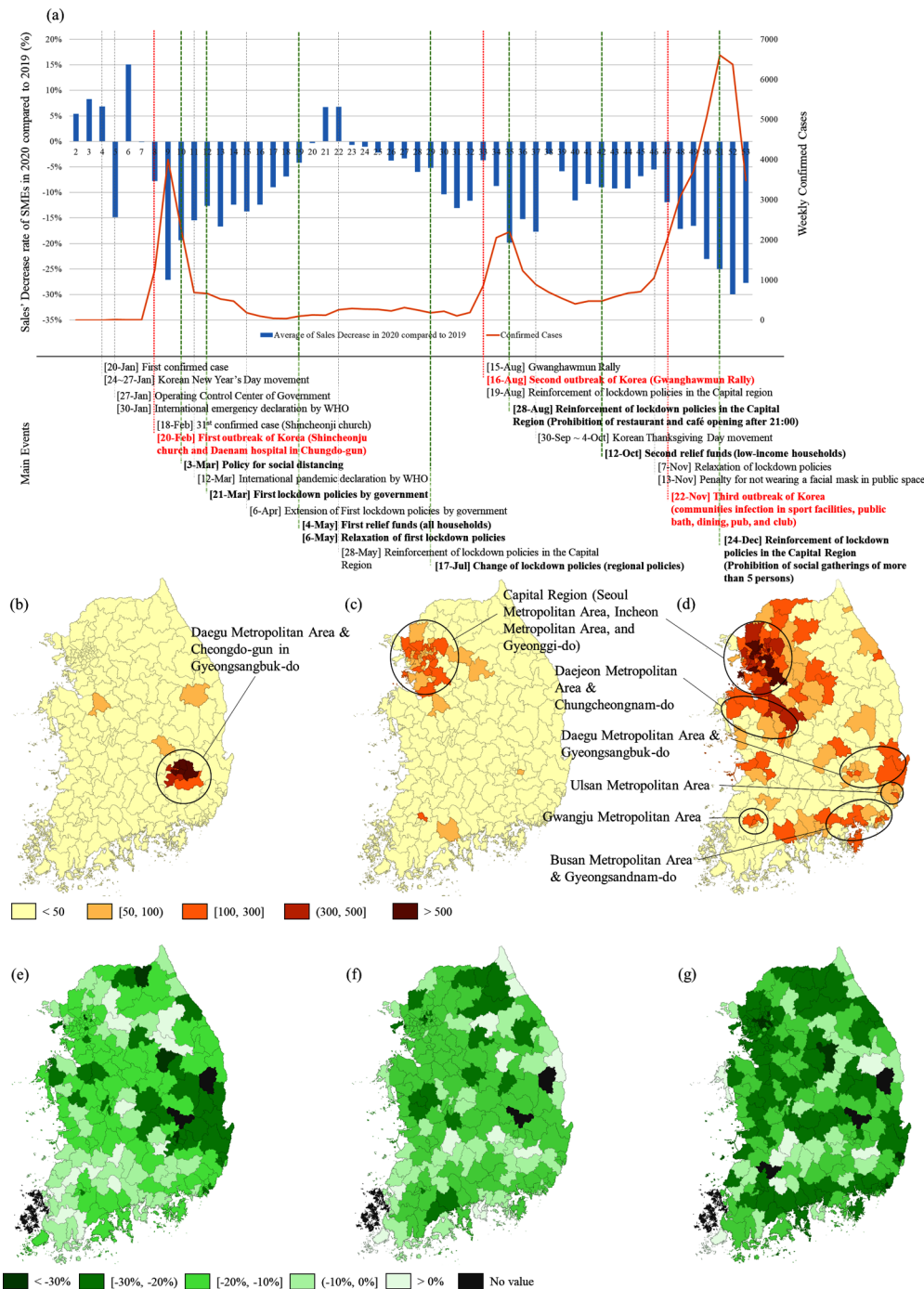


Figure 1. The regional COVID-19 outbreak patterns and micro, small, and medium enterprise (MSME) sales decrease across local municipalities in Korea. (a) Confirmed COVID-19 cases and average decrease in MSMEs' sales between weeks 2 and 53 (2020) in Korea; (b) confirmed cases during the first pandemic phase (weeks 7–11); (c) confirmed cases during the second pandemic phase (weeks 31–37); (d) confirmed cases during the third pandemic phase (weeks 46–53); (e) decrease rate of MSMEs' sales during the first pandemic phase; (f) decrease rate of MSMEs' sales during the second pandemic phase; and (g) decrease rate of MSMEs' sales during the third pandemic phase.

crisis patterns due to the pandemic and MSME crisis based on different phases. This study collected information on confirmed COVID-19 cases from week 2 to week 53 of 2020, as well as MSMEs' sales decrease data (comparison of 2020 data with 2019 data) (Korea Credit Data, 2021) (Figure 1).

The first outbreak's originating event was a group infection in the Daegu–Gyeongbuk region (Figure 1b). The first lockdown was implemented on March 21, 2020, and sales of MSMEs declined significantly in the outbreak area during this period (Figure 1e). The government implemented a first relief fund for all households on May 4, 2020, to alleviate the economic crisis caused by the first outbreak and relaxed the lockdown. From week 20 to week 29, the MSME sales decrease was followed by a recovery period where sales nearly reached 2019 levels. The second outbreak on August 15, 2020, linked to the Gwanghwamun Rally, occurred in areas around the capital region (Figure 1c). During the second outbreak, MSME sales decreased by 20% from 2019, but the regional pattern of MSME sales decrease differed from the COVID-19 outbreak pattern. The third outbreak, a small-scale group infection, occurred on November 22, 2020; it showed a regional pattern that differed from that of the first and second outbreaks. In all major cities, the number of confirmed cases increased sharply (Figure 1d); the MSME sales also decreased simultaneously because of the reinforced lockdown policies.

This research thus provided two novel findings. (1) The duration of the period during which the two outbreaks and economic shutdown effects appeared (after the reinforcement of the lockdown) has gradually increased; furthermore, the regional scope of the two event effects is also expanding. (2) The economic crisis occurred simultaneously with the COVID-19 crisis. The government may find it difficult to control the two crises simultaneously during a pandemic situation.

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References

- Chen Y-W, Ni L, Xu D-L, et al. (2021) Visualising regional disparities in the risk of COVID-19 at different phases of lockdown in England. *Environment and Planning A: Economy and Space*, 1–4. <https://doi.org/10.1177/0308518X20984165>
- Korea Credit Data (2021) Data portal. Available at: https://forum.cashnote.kr/data_portal (accessed 2 March 2021).
- Korea Disease Control and Prevention Agency (2020) Updates on COVID-19 in Republic of Korea. Available at: http://ncov.mohw.go.kr/en/tcmBoardView.do?brdId=12&brdGubun=125&dataGubun=&ncvContSeq=4552&contSeq=4552&board_id=&gubun= (accessed 2 March 2021).
- Zhai W, Liu M and Peng Z-R (2021) Social distancing and inequality in the United States amid COVID-19 outbreak. *Environment and Planning A: Economy and Space* 53(1): 3–5. <https://doi.org/10.1177/0308518X20932576>